

PHY 338k Lab Report 4

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1 Introduction

Labs 9–11 of PHY 338k: Electronic Techniques explores

2 Lab 9: Digital Electronics I: Logic Levels and Gates

2.1 Input logic signals and output signal measurement

To start, we built a logic voltage source using a DIP switch and pull-up resistors, then verified logic levels with the breadboard's logic monitor. To measure the logic thresholds, we lowered the input voltage until each channel's LED turned off. Our results are summarized in Table ??.

Our measured values were reasonably close to the specified 1.4V threshold given by the PB-503.

2.2 Quad NAND gate

We tested the 2-input NAND gate using both the 74HCT00N (CMOS) and 7400N (TTL) chips. Our measured outputs match what we would expect for a NAND gate, where it goes high for all inputs except 11.

As shown in Table ??, the CMOS gate output swings nearly the full 0V to 5V range, whereas the TTL gate shows a lower high output at around 3.4V. This is what we expect since the CMOS uses active-pull up that supports rail-to-rail output while TTL relies on bipolar transistors that limit high-level swing.

2.3 NAND gate circuits

2.3.1 NAND implementation of OR gate

Our experimental truth table in Table ?? confirms that the NAND gate circuit correctly implements the OR function.

2.3.2 A four input NAND gate circuit

This circuit implements the logic function $Y = (A + B) + (C \cdot D)$, which outputs 1 if either A or B is high, or if both C and D are high.

The truth table we obtained matches with the expected behavior for this circuit layout.

Figure ?? shows the output Y as high (1), which is consistent with the circuit's logic function.

2.4 CMOS logic gates from individual transistors

2.4.1 Inverter with passive pull-up

We built an inverter using an NMOS transistor and a passive pull-up resistor and drove it with a TTL square wave. Figure ?? shows our inverter producing a logical low output when the input is high, and vice versa.

At low frequency, the output followed the expected inversion with symmetric rise and fall times of $t_{\text{rise}} = 25.59\mu\text{s}$ and $t_{\text{fall}} = 24.57\mu\text{s}$, respectively. At high frequency, we obtained rise and fall times of $t_{\text{rise}} = 1.997\mu\text{s}$ and $t_{\text{fall}} = 192.8\text{ns}$, respectively. We observed the rising edge become noticeably slower while the falling edge remains sharp. This behavior is due to the RC charging delay from the pull-up resistor, which limits the speed of rising transitions.

2.4.2 Inverter with active pull-up

We built an inverter using NMOS and PMOS transistors in an active pull-up configuration and drove it with a TTL square wave. Again, our circuit correctly acted as an inverter since the output is low when the input is high and vice versa. The input (CH1, yellow) and output (CH2, blue) are shown in Figure ?? at both high and low frequencies.

At low frequency, the inverter behaved as expected with symmetrical rise and fall times of around $t_{\text{rise}} = 26.62\mu\text{s}$ and $t_{\text{fall}} = 25.60\mu\text{s}$, respectively. At high frequency, both transitions remained relatively sharp, with significantly faster rise and fall times of $t_{\text{rise}} = 177.1\text{ns}$ and $t_{\text{fall}} = 199.9\text{ns}$, respectively.

Unlike the passive pull-up configuration, the rising edge of the output does not break down at high frequencies since the PMOS transistor actively sources current to drive the output high while the NMOS pulls it low. In contrast, the passive pull-up relies on a resistor, which results in slower RC-limited charging. This improvement in switching speed explains why nearly all logic circuits use active pull-up design.

2.4.3 CMOS NAND gate constructed from individual transistors

NOTE: FINISH UP THIS PART LATER!!!

Show the circuit diagram you used with just the CD4007 pins and your external connections, and also another diagram showing the complete circuit with

the individual transistor connections.

Also show the CD4007 pins on the second diagram.

3 Lab 10: Flip-Flops and Sequential Logic

3.1 Flip-flops from NAND gates

3.1.1 Asynchronous SR flip-flop

We tested the SR flip-flop by applying a sequence of S and R inputs. Our circuit correctly set ($S = 0, R = 1$), reset ($S = 1, R = 0$), and held its state when both inputs were high. When both inputs were low ($S = 0, R = 0$), the circuit entered an invalid state with undefined outputs, which confirmed that this condition should be avoided. Our trial results are summarized in Table ??

3.1.2 Clocked SR flip-flop

Next, we tested the clocked SR flip-flop by applying various combinations of S , R , and CLK . When CLK was low, the output held its previous state regardless of inputs, and when it was high, the circuit performed set and reset functions as expected. There was an invalid state when both S and R were low during a high CLK input, which gave us undefined outputs. Our results are summarized in Table ??.

3.2 D-type flip-flop

3.2.1 D-type flip-flop operation

We tested the functionality of the D-type flip-flop by connecting a slide switch to the D input and using a debounced pushbutton switch to provide the clock signal. As expected, the output Q updated to match the value of D only on the rising edge of the clock, so we confirm that the flip-flop is positive-edge triggered.

We also tested the asynchronous inputs. Pulling the SET input low immediately forced Q high regardless of the clock or D input. Similarly, pulling the $RESET$ input low forced Q to low. Both inputs operated asynchronously and overrode the normal clock behavior. After releasing, the flip-flop went back to regular edge-triggered operation.

3.2.2 Flip-flop with feedback

Clock the circuit manually and observe the output.

Watch the Clock input and the Q output on an oscilloscope. What is the circuit doing?

Turn the frequency up and measure the propagation delay of the circuit.

Why is this relevant given the feedback connection from Q-bar to D?

We connected the D flip-flop in a feedback configuration by wiring $\bar{Q}$ to the D input. The output Q alternated between 0 and 1 with each clock pulse when we clocked it manually using a pushbutton. Thus, we confirm that our flip-flop circuit acts as a divide-by-2 frequency divider since Q toggles on every rising edge.

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3.2.3 Cascaded flip-flops

Using the function generator as the clock input for U1, observe and record the Q output of U2.

3.2.4 Shift register

Using the scope, observe the IN and Q output of the first flip-flop. Why the jitter?

Next, watch the output Q2 along with IN. Repeat for Q3 and Q4. What is the circuit doing?

4 Lab 11: Microcontroller Applications

4.1 Getting started with the Arduino IDE

You should now see the program running your Arduino, i.e. you should see the LED on the board blinking on and off

You should now see the LED blinking with your new time periods.

4.2 Connecting a display

Get the output text on the display

Change the program to output a few of your favorite words and verify that these new words are output on the display.

4.3 Measuring an analog voltage

Verify that the signal is the same as what you see on the oscilloscope

**aliasing - occurs due to periodic sampling of a periodic wave*

At about what frequency does the display fail to keep up with the input sine wave?

See if you can get the Arduino to record a faster sine wave. Verify that you can now see a faster sine wave on the display.

What's the fastest sine wave that you can faithfully record now?

4.4 Digital PID control loop

Open a serial plotter window with the baud rate to 9600 and view the resulting voltages

You should see the voltage drop to zero for a period of time, after which the feedback loop kicks in. What happens as the feedback tries to stabilize the voltage to the set-point?

Using the oscilloscope, observe the direct digital output at I/O pin 6, and also view the filtered output.

comment on how effective the filter is at removing the PWM from the output

Try changing the parameters k_p , k_i , and k_d to see what happens.

Record what happens for some representative set of parameters. What parameters cause the voltage to converge most quickly to the setpoint? Under what conditions is the voltage vs. time more stable? Less stable? Completely unstable?

Conclusion

Labs 9-11

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